

Timers

TinkerCAD Activity: “Timers”

This practice includes the following topics:

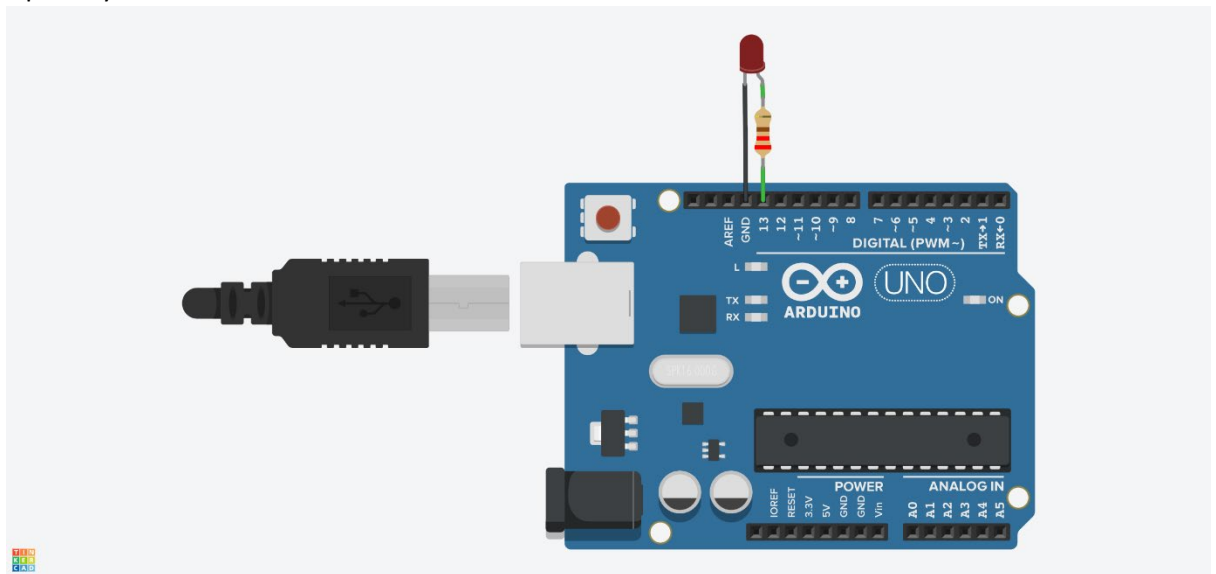
- Timer2 (Introduction & Basic Exercises).
- Overflow Interrupts using Timers.
- Compare Method.
- Timer1.
- Timer Interrupts with External Interrupts.

* In the following virtual exercises we will not take into account the “debouncing” method discussed in the previous lessons.

1. Tinker: “1. Timers - Blinking LED”

(Timers Overflow Interrupts – Timer2)

- Go to “Activities” and “Copy and Tinker” the associated task.
- Update your “submission file name”.

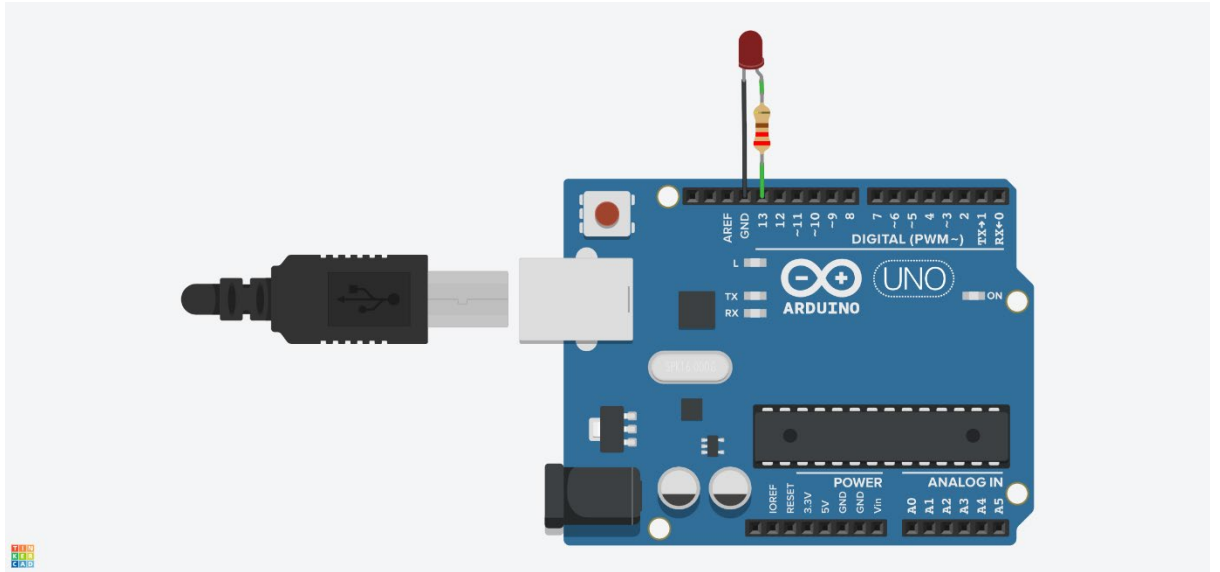


- Follow the instructions provided in TinkerCAD! and write a code that does the following:
 - Demonstrates the usage of **Timers Overflow Interrupts**.
 - Configuring Timer2 with a prescaler of 1024.
 - Every ~1second, Timer2 should generate an “Overflow Interrupt” and the associated ISR should toggle the LED.

Note: The clock is configured to be 16[MHz]. Take into account the “prescaler” as well as the fact that Timer2 is 8-bit counter.

2. Tinker: “2. Timers Blinking - Different Prescaler and Timing”

- a. Go to “Activities” and “Copy and Tinker” the associated task.
- b. Update your “submission file name”.

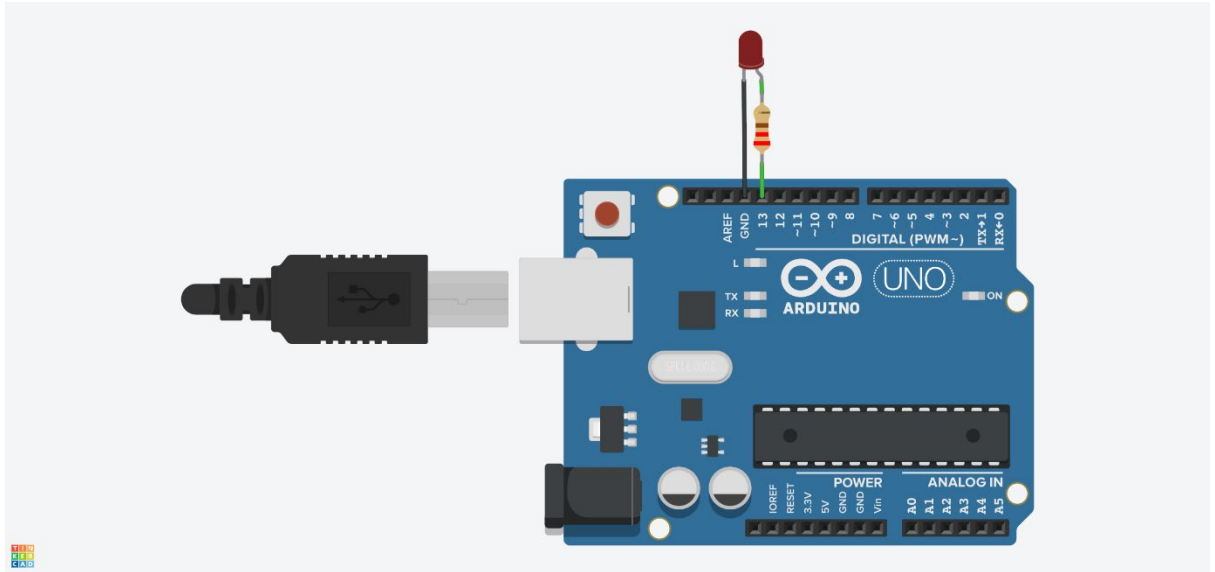


- c. Write a code that does the following:
 - i. Demonstrates the usage of **Timers Overflow Interrupts**.
 - ii. Configuring Timer2 with a prescaler of 128.
 - iii. Every ~0.5second, Timer2 should generate an “Overflow Interrupt” and the associated ISR should toggle the LED.

Note: The clock is configured to be 16[MHz]. Take into account the “prescaler” as well as the fact that Timer2 is 8-bit counter.

3. Tinker: “3. Three modes of blinking”

- a. Go to “Activities” and “Copy and Tinker” the associated task.
- b. Update your “submission file name”.



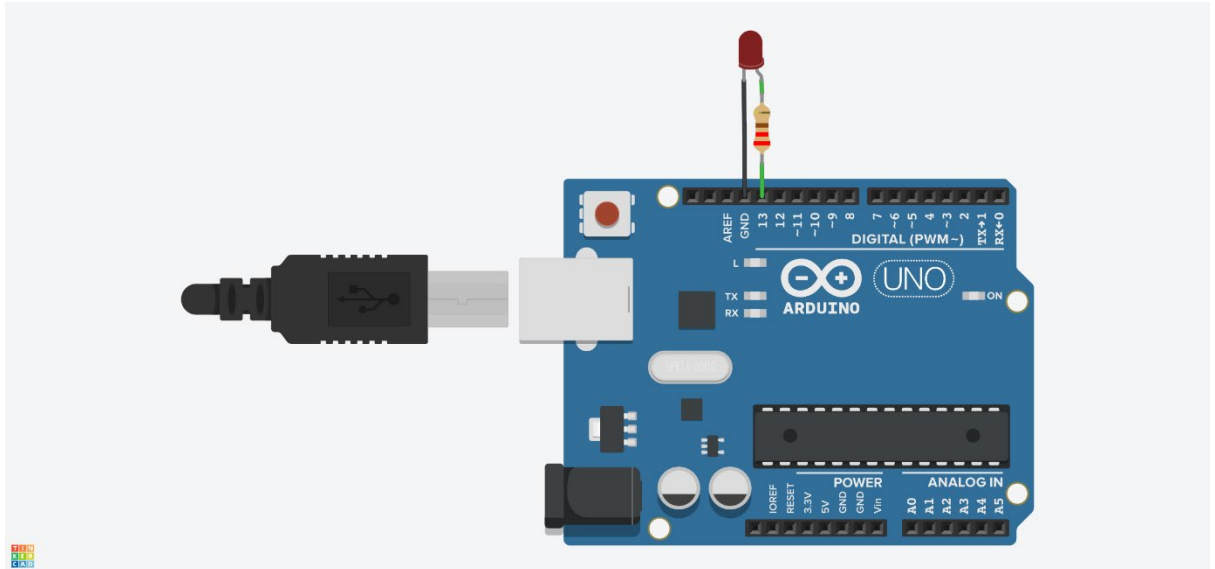
- c. Write a code that does the following:
 - i. Demonstrates the usage of **Timers Overflow Interrupts**.
 - ii. Configuring Timer2 with a prescaler of 1024.
 - iii. The program should be divided in three modes (that should be repeated every minute):
 1. Mode 1: The first 10 seconds the LED is going to blink at a 1 sec rate.
(1 second ON, 1 second OFF, 1 second ON, 1 second OFF, ...)
 2. Mode 2: The next 10 seconds the LED is going to blink at a 2 sec rate.
(2 seconds ON, 2 seconds OFF, 2 seconds ON, 2 seconds OFF, ...)
 3. Mode 3: The next 10 seconds the LED is going to blink at a 5 sec rate.
(5 seconds ON, 5 seconds OFF, 5 seconds ON, 5 seconds OFF)

And repeat to Mode 1.

Write from scratch your own solution.

4. Tinker: “4. Three modes of blinking (with MACROS)”

- a. Go to “Activities” and “Copy and Tinker” the associated task.
- b. Update your “submission file name”.



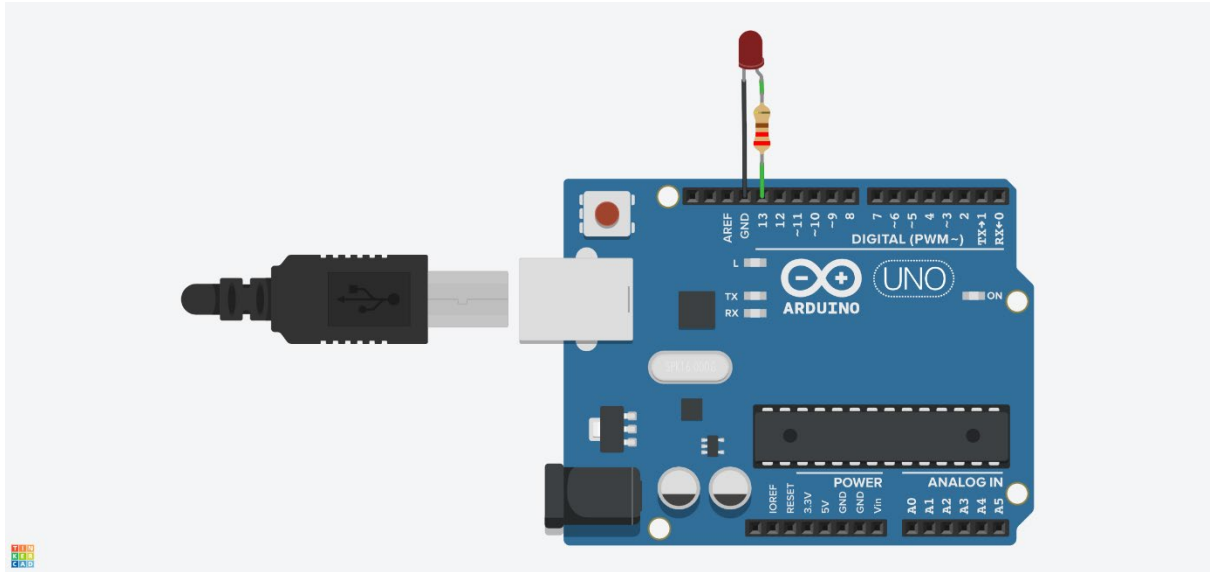
- c. Write a code that does the following (similar to previous exercise):
 - i. Demonstrates the usage of **Timers Overflow Interrupts**.
 - ii. Configuring Timer2 with a prescaler of 1024.
 - iii. The program should be divided in two modes:
 1. Mode 1: The first 10 seconds the LED is going to blink at a 1 sec rate.
 2. Mode 2: The next 10 seconds the LED is going to blink at a 2 sec rate.
 3. Mode 3: The next 10 seconds the LED is going to blink at a 5 sec rate.And repeat..
 - iv. **Your task is to practice the usage of macros**. Your program should include the following MACROS:

```
#define TIMER_PRESCALER_BITS _____  
#define TIMER_OVERFLOW_INTERRUPT_ENABLE_BIT _____  
#define TIMER_THRESHOLD_MODE_1 _____  
#define TIMER_THRESHOLD_MODE_2 _____  
#define TIMER_THRESHOLD_MODE_3 _____  
#define STATE_DELAY_MODE_1 _____  
#define STATE_DELAY_MODE_2 _____  
#define STATE_DELAY_MODE_3 _____
```

These macros should be used in your code.

5. Tinker: “5. Timers - Blinking LED - using Comparator”

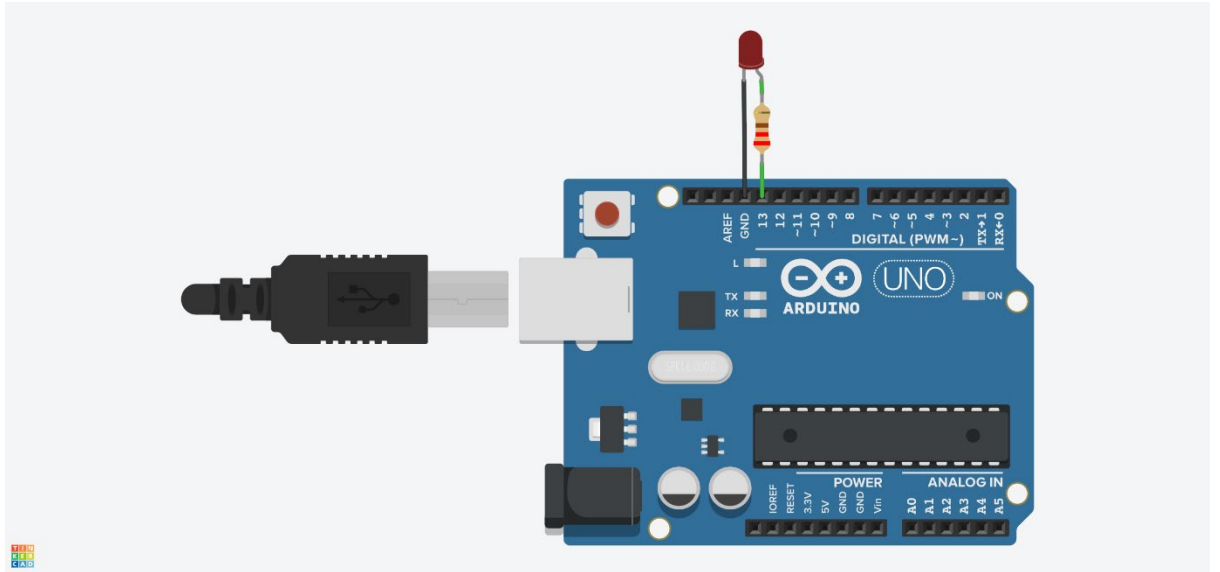
- a. Go to “Activities” and “Copy and Tinker” the associated task.
- b. Update your “submission file name”.



- c. Instead of working with Timers Overflow Interrupts your goal is to configure Timer2 to work in “Comparator Mode”
 - i. Set a prescaler of 1024.
 - ii. Configure relevant registers to provide an “exact” every second Interrupt using CTC Compare Mode. Calculate relevant values for the configurations.

6. Tinker: “6. Timers - Blinking LED - with Timer1”

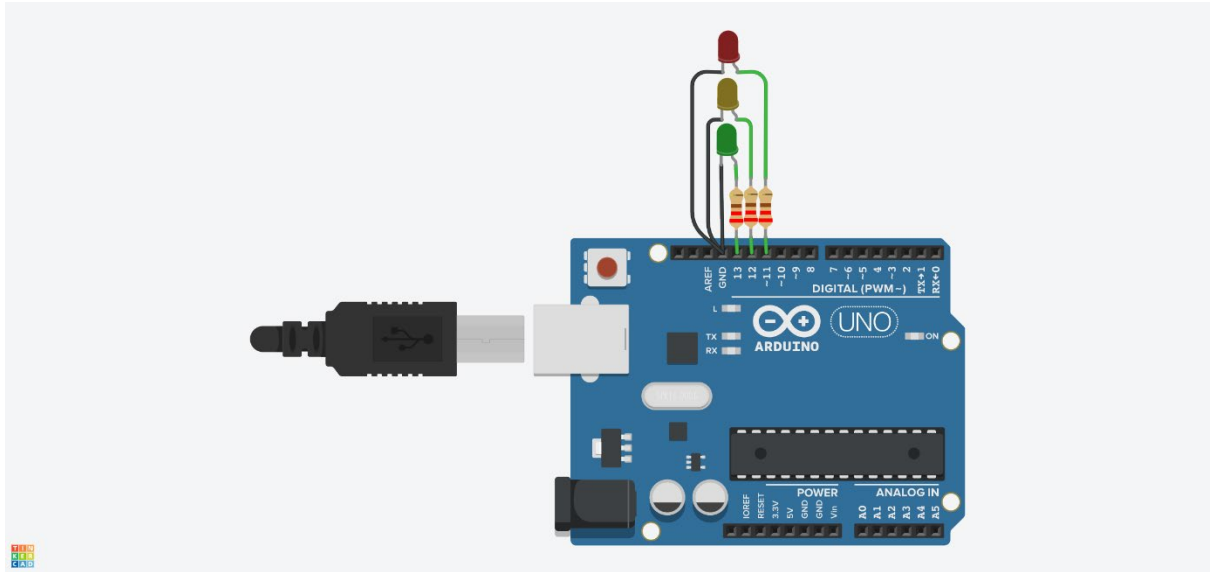
- Go to “Activities” and “Copy and Tinker” the associated task.
- Update your “submission file name”.



- Practice the working process with Timer1 – configure relevant registers, compare differences, etc.

7. Tinker: "7. Traffic_Light_Simulator - using Timer1"

- a. Go to "Activities" and "Copy and Tinker" the associated task.
- b. Update your "submission file name".



- c. This program should simulate the working of a Traffic Light.
 - i. Complete the attached code using Timer1 and Compare Mode.
 - ii. States explanation:
 1. A RED light, for 5 seconds.
 2. A RED light with YELLOW light, for 2 seconds.
 3. GREEN light only, for 5 seconds.
 4. A YELLOW light only, for 1 seconds.
 5. Repeat the process from the start (from the "First" state).

